Lagnesh Mahapatra

Perception & Autonomy Engineer
LAT Aerospace
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Examination	University	Institute	Year	CPI
Graduation	IIT Bombay	IIT Bombay	2021-2025	9.36
Intermediate/+2	CBSE	Mother's Public School	2021	98.60%
Matriculation	CBSE	ODM public School	2019	96.80%

SCHOLASTIC ACHIEVEMENTS

- Awarded the **'Silent Star'** recognition for consistent excellence and impactful contributions to **Skyroot** *[2025]*
- Graduated with Department rank **2** in the department out of **77 undergraduates** *[2025]*
- Received **Institute Academic Prize** for excellent academic performance for years **2022-23** and **2023-24** *[2024]*
- Secured **AP** grade for exceptional academic performance in **2** courses – Differential Equations II - MA 207 (**6** out of **655** students) and Compressible Fluid Mechanics - AE 236 (**1** out of **75** students) *[2023, 2022]*
- Secured an AIR of **2118** in **JEE Advanced** out of **0.15 million** aspirants *[2021]*
- Attained a percentile of **99.76** in **JEE Mains** among **0.9 million** candidates *[2021]*
- Selected for **NTSE Scholarship** offered by Ministry of HRD, Govt of India *[2019]*

RESEARCH EXPERIENCE

- **Robust Aeroelastic Response Estimation for Flexible Aircraft** by *Lagnesh Mahapatra* and *Omkar Halbe*, *IFAC-PapersOnLine*, 2024, DOI: 10.1016/j.ifacol.2024.05.017 *[2024]*

PROFESSIONAL EXPERIENCE

- **Perception & Autonomy Engineer** | ***LAT Aerospace, Gurgaon, India*** *(Feb'26–Present)*
  - Currently developing **EO/IR fused Object Detection and Multi-Object Tracking** pipelines using **YOLO-based architectures** for autonomous defense systems.
  - Designed **classical motion-based detection algorithms** for small/sub-pixel object detection using **Track-Before-Detect** techniques, and implemented **RF-DETR (Vision Transformer)** models for state-of-the-art detection performance.
  - Developed and validated a **GNSS-denied Visual SLAM pipeline** fusing **monocular camera, IMU, and rangefinder** data (**OpenVINS** backend), benchmarked against GPS ground truth.
  - Conducted a comprehensive **Monte Carlo sensitivity analysis** on the **Lat One v0.2 uSTOL demonstrator**, identifying flight-critical parameters and directly informing design and tuning decisions.
- **GNC Engineer** | ***Skyroot Aerospace, Hyderabad, India*** *(Dec'24–Jan'26)*
  - Contributing to the **RCS-based Control System Design** for the **Vikram-1 Launch Vehicle**.
  - Involved in **OILS** and **HILS Testing** of **ADCS Algorithms** for **CubeSat Missions**.
  - Developed a **Differential Evolution–based Global Optimizer** to obtain the **Optimal Descent Trajectory** for **Reusable Launch Vehicle Stages**.
  - Developed the complete **ADCS flight software stack** for **CubeSat Missions**.
- **GNC Intern** | ***Skyroot Aerospace, Hyderabad, India*** | ***Guide: Prithvi D. Sanghani*** *(May'24-July'24)*
  - Designed the **Attitude Control Autopilot** with **Flex Nozzle Control** for multiple flight phases.
  - Analyzed the **Controllability** of the **Launch Vehicle** across different flight stages and Conducted detailed **Sizing of Control Power Plants** for Flex Nozzle Control.
  - Developed a **6-DOF dynamic model** of a **CubeSat** in **C**, successfully implementing **Detumbling and Attitude Tracking controllers** to enhance satellite stability and orientation.
  - Conducted **Code Reviews**, **Static** and **Dynamic analysis** of **Digital Autopilot Software** using **LDRA**.
  - Authored Comprehensive **Software Design Documents** for **Digital Autopilot and Guidance Systems** as an integral part of the **Software Development Life Cycle**.
- **Trajectory Optimization Intern** | ***DRDL, DRDO, Hyderabad, India*** | ***Guide: Dr. Srijan Triparthi*** *(Dec'23-Jan'24)*
  - **Developed** and **Optimized** the **Reentry Trajectory** of a **Hypersonic Missile**, employing the **Gauss Pseudo Spectral Method** under **Multiple Constraints**, in accordance with contemporary **research** and **literature**.
  - **Implemented** the **Reentry Trajectory Optimization** strategies for a **Hypersonic Vehicle** from the paper "**Reentry Trajectory Optimization for Hypersonic Vehicle Satisfying Complex Constraints**."
  - Translated these strategies into **MATLAB code**, effectively addressing **complex constraints**.
- **Flight Dynamics Intern** | ***CSIR-NAL, Bangalore, India*** | ***Guide: Dr. Omkar Halbe*** *(Jun'23-July'23)*
  - **Integrated** a **Custom Attitude Controller** into **Ardupilot**, significantly streamlining the implementation of sophisticated control algorithms and enhancing its overall capabilities for advanced autonomous flight.
  - Worked extensively on both **Linear** and **Nonlinear Observer Design** for a **Hypersonic Aircraft**, contributing to substantially improved **state estimation** and enhanced **robust control** performance.

KEY PROJECTS

- **Guidance and Control of Spacecrafts** | *Bachelor's Thesis Project Part 1* | *Prof. Shashi Ranjan Kumar* *(Jan'24 - May'24)*
  - Implemented and conducted a deep analysis of the paper on **Predefined-Time Approximation-Free Attitude Constraint Control of Rigid Spacecraft**, thoroughly examining its methodologies and applications.
  - Enhanced **lunar landers** with **advanced control algorithms** for improved **precision** and **stability**.
- **Dynamic Stability Analysis of Learjet 24** | *Course Project* | *AE 305 - Prof. Arnab Maity* *(Apr'24 - May'24)*
  - Analyzed longitudinal and lateral dynamics of the Learjet 24 using linearized small perturbation equations and state-space modeling.
  - Derived transfer functions for longitudinal and lateral states, examining the influence of stability derivatives on eigenvalues and system responses.
  - Evaluated short-term and long-term aircraft responses to control inputs, highlighting discrepancies in damping values for short period and Dutch roll modes.
- **Attitude Control of Spacecrafts** | *Bachelor's Thesis Project Part 1* | *Prof. Shashi Ranjan Kumar* *(Aug'23 - Jan'24)*
  - Implemented the **'Fault-Tolerant Finite-Time Controller for Attitude Tracking of Rigid Spacecraft Using Intermediate Quaternion'** paper, gaining experience with advanced control techniques.
  - Investigated advanced methods for **Spacecraft Attitude** and Control, drawing insights from a study of **"Spacecraft Dynamics and Control"** by **Marcel J. Sidi**, to enhance understanding of spacecraft behavior.
  - Studied **"Applied Nonlinear Control"** by Slotine and Li to develop a solid understanding of **Nonlinear control techniques**, with emphasis on their practical applications in **Spacecraft Attitude and Control**.
- **PD Controller Design** | *Course Project* | *AE 308 - Prof. Arnab Maity* *(Nov'24 - Dec'24)*
  - Designed a PD controller to meet specified system performance criteria, including percent overshoot, settling time, gain margin, and phase margin.
  - Theoretically derived controller parameters and gains, validating the design through simulations to ensure desired specifications were achieved.
  - Analyzed Root Locus and ensured system stability, achieving a gain margin of 15 dB and phase margin of 79.5 deg degrees without any right-half plane poles.
- **Flutter Analysis of Aircraft Wings** | *SURP Project* | *Prof. Vivek Natarajan* *(Jul'23 - Dec'23)*
  - Researched advanced **flutter models**, obtaining **complex eigenfrequencies** to accurately define stability boundaries, and conducted a thorough comparative analysis for engineering model development.
  - Engaged in detailed **high aspect-ratio wing dynamics** modeling for **HALE aircrafts**, integrating these models with flutter analysis frameworks to ensure a comprehensive and robust analysis approach.
- **Control Theory** | *SOS Project* | *Mentors: Siddarth Kaushik and Dhrumil Lotiya* *(May'23 - Jul'23)*
  - Developed a solid and comprehensive foundation in **Control Theory**, including **feedback control** and **controller design**, through an in-depth study of "Control Systems Engineering" by Norman Nise.
- **Aircraft Engine Analysis** | *Course Project* | *AE 234 - Prof. Krishnendu Sinha* *(Jan'23 - Apr'23)*
  - Covered extensive literature on the design and development of the **Snecma Olympus 593** engine, which powered the iconic **Supersonic airliner Concorde**, and prepared a detailed report.
  - Conducted a thorough analysis of various engine components, from **Intake to Nozzle**, and applied learned concepts to meticulously calculate crucial **Performance Parameters** like **Thrust** and **Engine Efficiencies**.
  - Performed detailed **Python simulations** to evaluate the engine's performance under various conditions and thoroughly reviewed literature on potential design improvements for enhanced efficiency.

TECHNICAL SKILLS

- **Programming Languages**: Matlab, C, Arduino, C++, Python, LaTeX
- **Software**: FlightGear, Ardupilot, Simulink
- **Libraries**: NumPy, Pandas, Matplotlib, Seaborn
- **Ms Office Tools**: Excel, PowerPoint, Office

KEY COURSES UNDERTAKEN

- **Coursera and Udemy**: Introduction to Deep Learning and Neural Networks with Keras, Modern Robotics, Course 1: Foundations of Robot Motion, Introduction to the Internet of Things and Embedded Systems, Machine Learning with Python, Computational Fluid Dynamics Fundamentals Course, Computational Fluid Dynamics Fundamentals Course 2, Microcontroller Embedded C, Arduino for Beginners, Flight Instrument Essentials, An Entire MBA in 1 Course
- **Aerospace Engineering**: Modelling and Simulation, Navigation and Guidance, Plasma Physics for Engineers, Flight Dynamics and Control, Aircraft Design, Flight Mechanics, Control Systems Laboratory, Aerodynamics Laboratory, Aircraft Propulsion Laboratory, Aerodynamics, Control Theory, Vibration and Structural Dynamics, Aerospace Propulsion, Aircraft Structures Laboratory, Aircraft Propulsion, Compressible Fluid Mechanics, Aerospace Structural Mechanics, Spaceflight Mechanics, Aerospace Measurements Laboratory, Thermodynamics and Propulsion, Solid Mechanics, Incompressible Fluid Mechanics, Introduction to Aerospace Engineering, Data Analysis and Interpretation
- **Mathematics**: Linear Algebra, Multi-variable Calculus, Ordinary Differential Equations, Partial Differential Equations, Introduction to Numerical Analysis, Mathematical Structures For Control
- **Advanced Electives**: High Performance Scientific Computing, Stress Analysis, Finite Element and Boundary Element Methods